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DESIGNING A 2D COLOR BARCODE

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***Abstract*— The barcode technology has undergone many dramatic changes during the past few years. This is due to the increased need for encoding more data into the barcode. One of the latest developments in the barcode technology is color barcode, developed by Microsoft's High Capacity Color Barcode (HCCB). With four or eight colors, color barcodes are now used extensively in certain areas such as airports, print advertising and mobile phones compared to the 2D black and white barcodes such as Quick Response (QR) code and data matrix. This paper provides an overview of six papers related to color barcode and later introduces an algorithm developed to encode data using binarization and grouping of bits to form a color barcode and also to decode data from color barcode.**

I. INTRODUCTION

BARCODE is a representation of digital data that are encoded in it and can be read using an optical machine. Due to its advantage over human performance in accuracy, speed and so on, barcode technology is used extensively. Once the traditional 1D barcode was developed, people started to think of developing 2D barcodes. 2D barcodes can be formed using simple geometrical shapes such as square, circle, hexagon, triangle and so on, rather than using adjacent parallel lines. Moreover, the need to maximize data capacity of the barcode, led to the development of 2D barcodes such as Quick response (QR) codes and color barcodes. A QR code is a graphical representation of data in vertical and horizontal positions over a fixed space. For example, the Microsoft's High Capacity Color Barcode (HCCB) or the tag barcode (Fig. 1). has the data density of 16,000 bits per square inch which is three times larger than data density in QR barcodes [1]. Nowadays, application of QR code can be seen in airport boarding pass, retail shops, company logo and so on [1].

The main challenge of the color barcode technology is the intensity and depth of the colors used in barcode. In barcode, colors are not used for aesthetic purpose only, but rather they increase data capacity. The drawbacks in using the color can occur only during the print and scanning time, which depend on the paper and color quality. The six papers reviewed in this manuscript are mainly related to the challenges such as color blur, intensity of color variations and complex color classifiers existing in the barcode during the decoding time.

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In this paper we discuss the literature survey and an implementation algorithm for encoding the data into the barcode. The strategy used is converting the information to binary bits and assigning colors to the corresponding binary pairs. By this method it is possible to maximize the data capacity. When decoding, the reverse of the encoding is done to retrieve the data from the barcode.

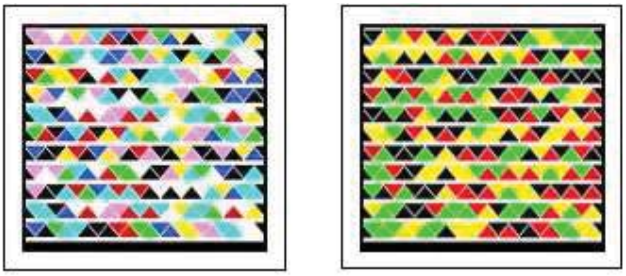


Fig. 1. The Microsoft tag Barcodes [2]

II. LITERATURE REVIEW

A literature survey on 2D color barcode brought about a new development to the barcode technology. The aim was to maximize the data density while reducing error. Using four and eight colors a new color barcode was developed that could store more data than QR code (Fig.1). The reference color pallet was used for decoding whereas the black and white borders are used for the orientation and detection process. The drawback of the system was its inability to detect and align the color barcode [1].

Querini et al. [1] used a new formation of barcode which was built on the QR code platform[1], named High Capacity Colored 2-Dimentional (HCC2D). The technique was similar to that of HCCB. The optimization problem was to maximize the data density and was subjected to certain equality constraints such as preserving the same robustness and error correction properties as the QR code. The reference color palette was used for decoding color barcode during the scan process. The paper discussed the experiments carried out using desktop computers and mobile systems. The pro for this method is that it maintains the same robustness and error correction of the QR code. However, the success rate of this technique depends on the print resolution of the colors.

Keng and colleagues [3] presents a new finder pattern and recognizing algorithm for decoding barcode. The optimization function is to maximize data capacity which is

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subject to an equality constraint that is utilizing symbol space with more colors. While the technique was successful in decoding the barcode at different light intensities, the black and brown borders often created problems during decoding and the finder pattern created minute errors at low pixels.

Bagherinia and Manduchi [4] formed an algorithm to decode the barcode without using the color palette. The optimization algorithm used is joint color representation in which color changes form a low-dimensional matrix. This technique can maximize the information rate with less probability of error. The challenge of this technique is that the blur due to mixing of the colors of nearby patches causes errors. There are also problems due to color drift during printing [4].

Querini and Italiano [5] meant to analyze the most effective algorithm for decoding HCC2D barcodes. The experiments were conducted with five different algorithmic methods and the goal was to find the best suitable redundancy rate. From these algorithms, K-mean clustering method gave the best outcomes with the least mean and standard deviation. Although the authors were able to develop a suitable color classifier for converting analog barcode cells to digital bit streams, the more complex color classifiers showed low accuracy in classifying the color barcode cells [5].

Bulan and Sharma [6] proposed an Expectation Maximization (EM) algorithm attempted to improve the color barcode during color interference. They were able to cancel the color interferences and also improved bit error rate performance of the secondary colors such as cyan, magenta and yellow. However, they were not able to improve the bit error rate of cyan using their method.

Simske and colleagues [7] attempted to maximize the security of the barcode and the payload density. They used eight colors to form color tiles in the color barcode. In order to maximize the security function of the color barcode two algorithms were performed for the authentication of the color barcode: The RGB Distance Approach and Hue approach[6]. The Hue distance approach gave better results. Also, for better restoration, saturation equalization approach was used to maximize the payload density (PD). The equation of PD is [7]:

$$PD = \ln 6/\ln 2 * \text{number of PI} / 8 \quad (1)$$

where number of PI is the number of payload blocks at 1 inch^2 [7]. Using this using authentication algorithm the payload density was maximized and better security was achieved. However, whether the algorithm can be applied in mobile conditions was not researched.

III. THEORY

The main features of the color barcodes are the color models which can be divided into two categories: RGB and CMY(K) models (Fig. 2). These two can be used to constitute and give a variety of color intensity for developing color barcodes. Usually the RGB model is formed by the spectral combination of two colors to create a set of secondary colors. On the other hand, in the CMY(K) model, also known as subtractive primaries, the white color disappears by subtracting the three colors such as Cyan, Magenta and Yellow [8]. However, these three colors form a new color called the Key color that is black. These colors can vary in intensity to form 10, 12 or more color sets. As the intensity of the colors varies, the color barcode becomes more complex.

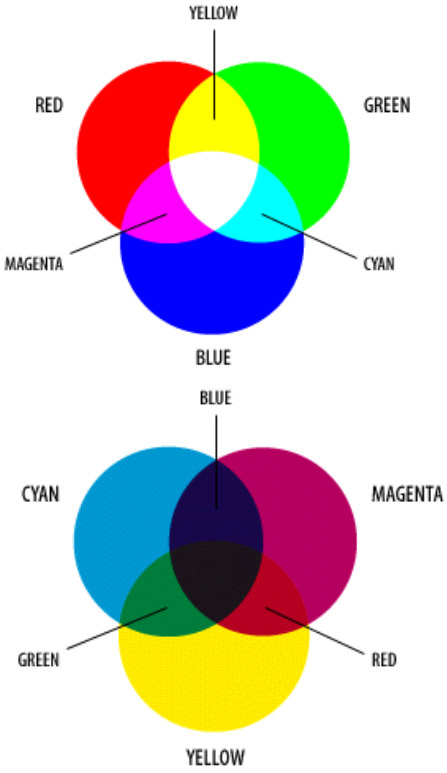


Fig. 2. The RGB (top) and CMY(K) (bottom) models

The colors not only depend on the surface spectrum but also on the unknown spectrum features such as color drift, color fade and so on. The unknown spectrum features are removed by Clustering technique and the Euclidean distance method.

In clustering technique, some clusters are identified for classifying the color cells using the mean shift value and then assigning the corresponding colors from the reference color set which is already contained in the barcode.

In the Euclidean distance method, the distance is calculated between reference color sample and encoded color sample. The minimum distance indicates maximum similarity.

Nonetheless these methods are good for small size barcodes with limited colors. As the color combinations increase, they become more complex and new methods need to be adopted. The density of data encoded always depends on the number of colors used [1]:

$$bpm = \log_2 N \quad (2)$$

where bpm is the number of bits used for encoding data to a single square module and N is the number of reference colors taken. For example, if the number of colors taken for the barcode is 8, then bpm=3. Hence, if the number of colors is increased then the data that can be encoded also increase.

In order to scan data from the barcode, the reference colors need to be known in advance. Depending on the number of colors, the data density can increase. Data density can be increased in two ways: 1) using more data cells within the symbol space, 2) using more colors within the symbol space. Nowadays, color barcode decoding can be performed using mobile phones and this can cause new challenges in computer vision. The challenges are due to the color balance difference for different types of barcode readers and variation in the geometry of the barcode [1]. These challenges arise during the print and scanning process of the barcode. For example, usually cyan, magenta and yellow are the color channels used for printing whereas red, green and yellow are the color channels used for color images. To overcome such challenges, high quality image paper need to be used.

Orientation correction and color identification are the other challenges in barcode technology. In order to correct the orientation of a barcode, appropriate borders need to be provided. A simple way to correct the orientation of a barcode is rotating and correcting of border position according to the tangential angle, i.e., the angle between the opposite side and adjacent side of the barcode border. Also, in the color identification process, the barcode should be tested at different lighting conditions such as in shadow or under ordinary light conditions. A high resolution mobile phone can be used to ensure a barcode reader can read the data under these challenging conditions.

IV. IMPLEMENTATION

The barcode formulation with the use of eight colors is a challenging task, especially for desktop and mobile applications. In order to robustly encode and decode the data into and from a barcode, optimization algorithm formulation is necessary. In this paper, some design constraints in developing 2D color barcodes are considered:

Inequality constraint: text length of the data≥13,

Equality constraints: color square blocks=3 bits per module

- the 2D color barcode should be flexible according to the data capacity of the file.
- the data capacity of the 2D color barcode should be maximized using eight colors such as red, green, blue, cyan, magenta, yellow, white and black.

The key process in encoding is binarization of the data file and grouping of the bits. Once the raw barcode is formed, a border is provided mainly to correct the orientation. All information encoded to the barcode are eventually decoded from it. For example, using 2D color barcodes in boarding passes allows faster retrieval of passengers' information and flight schedules at airports or railway stations. Using 2D color barcodes saves time and prevents human errors. Furthermore, 2D color barcodes have higher data capacity than black and white 2D barcodes. The data file may contain any type of characters, symbols or numerals as specified in the ASCII lookup tables [9]. The figure 2 and figure 3 shows the block diagram representation of data encoding and data decoding.

The following steps are taken for encoding data to the 2D color barcode:

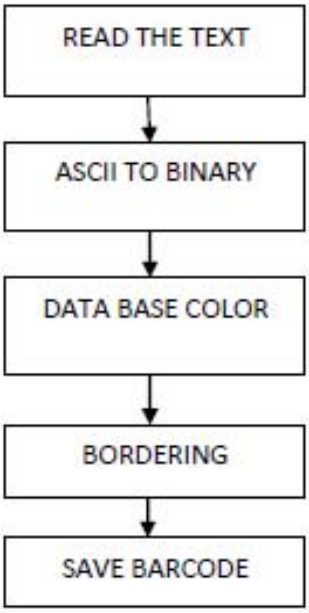


Fig. 3. Data Encoding

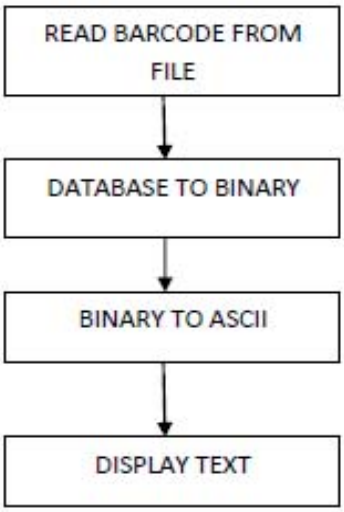


Fig. 4. Data Decoding

ASCII to Binary form

As for computing, binarizing starts as soon as the text is taken for character coding which contains a set of standards, used to represent a collection of characters for coding [10]. The data is converted to bits according to the decimal values

in ASCII lookup tables. The ASCII lookup table is used to represent data as a set of bytes and the bytes ranges from 0x00 to 0xFF. The lookup table contains English alphabets (both upper and lower cases), numerals, punctuations, control characters and so on. The alphabets are in the range of values between 0x21 and 0x7E, space character as 0x20, tab as 0x09, carriage return as 0x0A and line fed as 0x0D [11]. In general, character coding is used for computation, transmission and storage of text. Once the ASCII characters are coded, they are clustered together to form a set of binary information and each cluster consists of eight bits of information. The code page in a barcode is a form of data processing that represents digital data as a byte wise sequence. The byte can have any value [11] according to the characters in the data.

Color Blocks and Bordering

Figure 5 shows the eight color formation. The color assigning is based on clustering of three bits which correspond to eight colors, such as black, white, red, green, blue, cyan, magenta and yellow. Each of the bits is represented in the matrix and superimposed to form the aforementioned colors. Hence the raw barcode is formed and a thick uniform black border is built to correct the orientation as explained earlier.

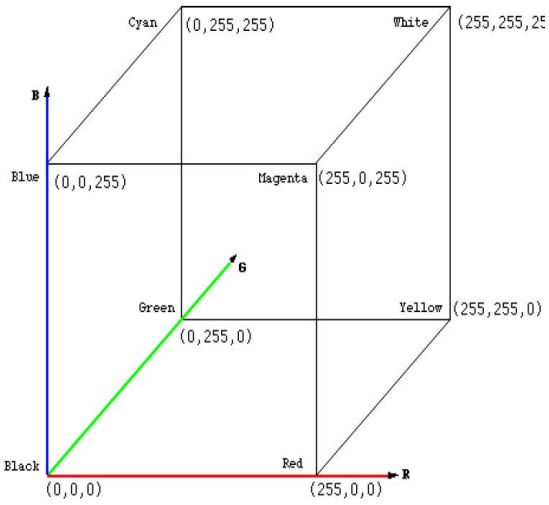


Fig. 5. Eight color formulation [7]

The following steps are taken for decoding the 2D color barcode:

Calculating the total data cell

Once the raw barcode is formed, the number of data cells is calculated prior to the barcode scanning. Each color block is considered during the calculation of data cells.

Retrieving color cells

In this step, it is necessary to cluster the bits into a matrix and convert them back to their corresponding color models.

Binary to text form

This is the reverse process of encoding where all the binary numbers are grouped to get the original text. The process is based on the color values obtained in the previous step. Once bits are grouped, the corresponding encoded information is retrieved. Usually, most computers store data in eight bit representation. The files are stored in ASCII format and have all possible eight bit values [12]. However, if non-ASCII characters appear in the text, the text will not be decoded properly.

For all these conditions, the decoding is done for the desktop computers, and the errors occurring when mobile phones are used are not considered.

V. RESULTS AND DISCUSSIONS

The 2D color barcode was developed with eight colors including the primary colors, secondary colors, and black and white. The developed barcode showed similar results as the one developed by Keng et al. [2]; the only difference between the two is that in the barcode developed in this study the borders are modified. The barcode developed in this study has higher data capacity (70% more data capacity than black and white barcodes) with each square module in the data set requiring only 3 bits. Since the developed barcode has thick black borders, it works well in ideal conditions where other color distortions do not affect it. The importance of border is that it provides information regarding the position from which decoding should be started.

Furthermore, the developed barcode border is not suitable for mobile decoding. To be effective in mobile applications, the barcode should have guide blocks and bars [2] and reference cell colors. The developed barcode works well for desktop systems even if the border is black or white. Figure 6 shows the developed barcode with black border and eight colors.

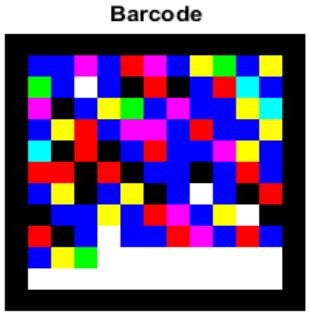


Fig. 6. Developed Color Barcode

VI. CONCLUSION AND FUTURE WORK

The developed 2D color barcode has many advantages, such as high data capacity, data security and compression, over the traditional black and white barcodes. The development of color barcodes is still challenging since the intensity and depth of colors vary during printing and scanning processes. By using eight colors, it became easier to develop any kind of barcode which is flexible according to the amount of

information available. This study was successful in achieving its goal of developing a 2D color barcode that can encode and decode information on a desktop computer. Future studies should focus on developing barcode borders with guide blocks and guide bars [2], rather than a uniform black border. This will help decoding data in a better and more orderly way.

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